



Pathfinding

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Difficulty Level : Medium

Description

Amina wants to travel from **point A** to **point B** in a grid-like city. The city is represented as a matrix where:

- **1** represents an open road where she can move.
- **0** represents a blocked road that she cannot pass.

Amina can only move **up, down, left, or right**. The challenge is to find the shortest possible path.

Problem Statement

Given a rows x cols matrix of 0s and 1s, find the shortest path from point A to point B.

- She can move in four directions (up, down, left, right).
- If reaching the destination is impossible, return an empty array.

Constraints

- $1 \leq \text{rows}, \text{cols} \leq 100$
- The starting and ending points always have the value 1.

Examples

Example 1

Input	Output
4 4 1110 0011 0110 0011 1 1 4 4	(1,1), (1,2), (1,3), (2,3), (3,3), (4,3), (4,4)

Example 2

Input	Output
3 3 100 001 111 1 1 3 3	[] # No path available